

∞ -categories of dimensions ≤ 1

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GK-dimension of algebras

- Let k be a field. For a finitely generated, connected graded k -algebra $A = TV/(R)$, one can associate a growth function

$$d_V(n) = \dim A_{\leq n}.$$

- In particular, there is the Gelfand–Kirillov dimension

$$\text{GKdim } A = \limsup \log_n \dim(A_{\leq n}).$$

Studies on the GK-dimension

- GK-dimension is an important invariant that is closely related to the Hilbert series (Stephenson and Zhang 1997).
- People have been classifying affine Noetherian (infinite dimensional) Hopf algebra of small GK-dimensions, along the study of non-commutative algebraic geometry and representation of quantum groups (Brown and Goodearl 1997; Lu et al. 2007; Wu et al. 2016; Andruskiewitsch and Sanmarco 2021).

Main questions

1. What is the GK-dimension for simplicial sets?
2. How to characterize/classify simplicial sets or ∞ -categories of small GK-dimensions?

Simplicial sets

Let Δ be the category whose objects are linearly ordered sets $[n] = \{0 < 1 < \cdots < n\}$ and morphisms are non-decreasing maps.

Definition

A simplicial set X is a functor $\Delta^{op} \rightarrow \mathbf{Sets}$.

Definition

Equivalently, a simplicial set X is a collection of sets $X_n = X([n])$ for $n \geq 0$ with face maps d_i and degeneracy maps s_i for $0 \leq i \leq n$ (except that there are no d_0 on X_0) such that

such that the following simplicial identities hold

$$\begin{aligned} d_i d_j &= d_{j-1} d_i & i < j \\ s_i s_j &= s_{j+1} s_i & i \leq j \\ d_i s_j &= \begin{cases} s_{j-1} d_i & i < j \\ \text{id} & i = j, j+1 \\ s_j d_{i-1} & i > j+1. \end{cases} \end{aligned}$$

Definition

A simplicial map from simplicial set X to Y is a natural transformation $X \rightarrow Y$. Equivalently, it is a collection of maps that commute with the face and degeneracy maps.

Examples of simplicial sets

Example

$\Delta^n = \text{Hom}_\Delta(-, [n])$ is a simplicial set.

Definition

The i -th horn of Δ^n is Λ_i^n where

$$(\Lambda_i^n)_k = \{f \in \text{Hom}_\Delta([k], [n]) : \text{im } f \cup \{i\} \neq [n]\}.$$

When $0 < i < n$, Λ_i^n is called an inner horn.

The boundary of Δ^n is $\partial\Delta^n$ where

$$(\partial\Delta^n)_k = \{f \in \text{Hom}_\Delta([k], [n]) : f \text{ not surjective}\}.$$

∞ -categories

Definition

A simplicial set X is an ∞ -category if, for all $n > 2$ and $0 < i < n$, every simplicial map $F : \Lambda_i^n \rightarrow X$ can be extended to a simplicial map $\tilde{F} : \Delta^n \rightarrow X$ such that $F = \tilde{F} \circ \iota$ where $\iota : \Lambda_i^n \hookrightarrow \Delta^n$ is the inclusion. That is, the following diagram commutes.

$$\begin{array}{ccc}
 \Lambda_i^n & \xrightarrow{\forall F} & X \\
 \downarrow \iota & \nearrow \exists \tilde{F} & \\
 \Delta^n & &
 \end{array}$$

Examples of ∞ -categories

Example

Δ^n is an ∞ -category but Λ_i^n and $\partial\Delta^n$ are not.

Conjecture

No quotients of Δ^n are ∞ -categories except for Δ^m for $m \leq n$.

Example (Goerss and Jardine 2009, p.13)

Any simplicial group is an ∞ -category.

GK-dimension of simplicial sets

Definition

A simplicial set X is locally finite if X_n is a finite set for all $n \geq 0$. Let X be a locally finite simplicial set. The GK-dimension of X is

$$\mathrm{GKdim} X := \limsup \log_n |X_{\leq n}|.$$

Example

$\mathrm{GKdim} \Delta^d = d + 1$ because $|\Delta_n^d| = \binom{n+1+d}{d}$.

$\mathrm{GKdim} \partial\Delta^d = d$ because $|\partial\Delta_n^d| = \binom{n+1+d}{d} - \binom{n}{d}$ for $n \geq d$.

Dimension of simplicial sets

Definition

Let X be a simplicial set. An element x in X_n is called an n -simplex. An n -simplex x is degenerate if $x = s_i(y)$ for some $y \in X_{n-1}$ and $0 \leq i \leq n-1$. Otherwise, x is non-degenerate.

Definition

The dimension of simplicial set X is

$$\dim X := \sup\{n : X_n \text{ contains an non-degenerate simplex}\}.$$

Example

$$\dim \Delta^d = d \text{ and } \dim \partial \Delta^d = d - 1.$$

Proposition

Let X be a locally finite simplicial set. Then

$$\mathrm{GKdim} X = \dim X + 1.$$

The idea is to decompose the simplicial set into disjoint unions where the size of each component can be easily computed. We need the following well-known Eilenberg-Zilber Lemma.

Lemma (Gabriel and Zisman 1967, p.26-27)

Let X be a simplicial set and $x \in X_n$ be degenerate. Then

$$x = s_{j_r} \cdots s_{j_1}(\tau)$$

for a non-degenerate τ and $n > s_{j_r} > \cdots > j_1 \geq 0$. Moreover, the choice of $\tau, s_{j_1}, \dots, s_{j_r}$ is unique.

Proof of the Proposition.

For each non-degenerate (say an n -simplex) x , define

$$\widetilde{\Delta}_t^x = \{y \in X_t : y = s_{i_1} \cdots s_{i_{t-n}}(x) \text{ for } t > i_1 > \cdots > i_{t-n} \geq 0\}.$$

Let $\widetilde{\Delta}^x = \coprod_{t \geq n} \widetilde{\Delta}_t^x$ and $\mathcal{NG}(X)$ be all non-degenerate simplices.

By Eilenberg-Zilber Lemma,

$$X = \coprod_{x \in \mathcal{NG}(X)} \widetilde{\Delta}^x.$$

Each component $|\widetilde{\Delta}_t^x| = \binom{t}{t-n} = \binom{t}{n}$ has polynomial growth. $\square$

Characterization of simplicial sets of $\dim \leq 1$

Theorem (Lurie 2018, Proposition 1.1.6.9)

The category of simplicial sets of dimension ≤ 1 is equivalent to the category of quivers.

The equivalence is given by $\Gamma : \mathbf{sSets}_{\dim \leq 1} \rightarrow \mathbf{Quivers}$ where ΓX is the quiver with 0-simplices x as vertices and non-degenerate τ 1-simplices as edges from $d_1(\tau)$ to $d_0(\tau)$. The inverse functor S takes a quiver Q to the pushout of the diagram

$$\begin{array}{ccc} \coprod_{e \in E(Q)} \partial \Delta^1 & \hookrightarrow & \coprod_{e \in E(Q)} \Delta^1 \\ \downarrow (s,t) & & \\ \coprod_{v \in V(Q)} \Delta^0 & & \end{array}$$

Characterization of ∞ -categories of $\dim \leq 1$

Definition

Let $\mathcal{C}$ be a small category. The nerve of $\mathcal{C}$ is a simplicial set $\mathcal{N}(\mathcal{C})$ where $\mathcal{N}(\mathcal{C})_n$ contains n -tuples of composable morphisms $(f_1, \dots, f_n)$ in $\mathcal{C}$.

Theorem

Under the equivalence given by functors Γ and S ,

- 1. The full subcategory of ∞ -categories of dimension ≤ 1 is equivalent to the full subcategory of quivers that has no paths of length 2.*
- 2. In particular, $S(Q) \cong \mathcal{N}(Q)$ for all quivers Q with no paths of length 2.*

Proposition

Let X be an ∞ -category of dimension ≤ 1 . Then ΓX has no paths of length 2.

Proof.

Any path of length 2, say $e_2 e_1$, determines a simplicial map $F : \Lambda_1^2 \rightarrow X$, and e_1, e_2 correspond to non-degenerate 1-simplices.

As X is an ∞ -category, the simplicial map extends to $\tilde{F} : \Delta^2 \rightarrow X$.

By Yoneda Lemma, the simplicial map $\tilde{F}$ corresponds to a 2-simplex $\tau \in X_2$ where $d_0(\tau) = e_2, d_2(\tau) = e_1$.

This means τ has two non-adjacent non-degenerate faces.

But τ is degenerate and can only have adjacent non-degenerate faces. So no path of length 2 exists. □

Proposition

Suppose Q is a quiver that has no path of length 2. Then the nerve $\mathcal{N}(Q)$ is isomorphic to $S(Q)$.

Proof.

Clearly $\mathcal{N}(Q)$ lies in the category of simplicial set of $\dim \leq 1$.

By Yoneda Lemma, there are simplicial maps $\psi_e : \Delta^1 \rightarrow \mathcal{N}(Q)$ corresponds to $e \in \mathcal{N}(Q)_1$ and $\phi_v : \Delta^0 \rightarrow \mathcal{N}(Q)$ corresponds to $v \in \mathcal{N}(Q)_0$. Taking coproducts, we get

$$\begin{aligned}\Psi &= \coprod_{e \in E(Q)} \psi_e : \coprod_{e \in E(Q)} \Delta^1 \rightarrow \mathcal{N}(Q) \\ \Phi &= \coprod_{v \in V(Q)} \phi_v : \coprod_{v \in V(Q)} \Delta^0 \rightarrow \mathcal{N}(Q)\end{aligned}$$

Then $(\mathcal{N}(Q), \Phi, \Psi)$ satisfies the universal property of pushout. $\square$

Simplicial homotopy equivalence

Definition

Let $f, g : X \rightarrow Y$ be simplicial maps. A simplicial homotopy from f to g is a simplicial map $h : X \times \Delta^1 \rightarrow Y$ such that $h|_{X \times 0} = f$ and $h|_{X \times 1} = g$. We denote $f \sim g$.

Definition

A simplicial map $f : X \rightarrow Y$ is a simplicial homotopy equivalence if there is a simplicial map $g : Y \rightarrow X$ such that $fg \sim \text{id}_Y$ and $gf \sim \text{id}_X$.

Proposition

Let X, Y be ∞ -categories of dimension ≤ 1 . Then X and Y are simplicial homotopic equivalent iff ΓX is isomorphic to ΓY .

Future questions

1. How to characterize weak simplicial homotopy equivalence between ∞ -categories of $\dim \leq 1$?
2. How to characterize simplicial sets of $\dim \leq 1$ up to (weak) simplicial homotopy equivalence?
3. What models simplicial sets of $\dim \leq 2$?
4. What should be the corresponding notion of homotopy in the category of quivers under the equivalence?
5. Is there a good notion of representations of ∞ -category?

Thank you!

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